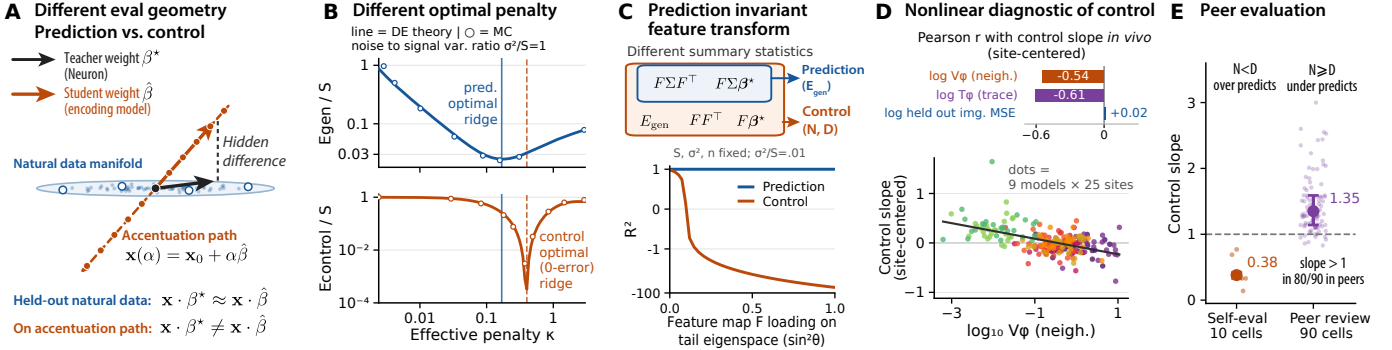

A
Different eval geometry
Prediction vs. control.
Teacher weight $\beta^\star$
(Neuron)
Student weight $\hat{\beta}$
(encoding model)
Natural data manifold
Hidden
difference
Accentuation path
$\mathbf{x}(\alpha) = \mathbf{x}_0 + \alpha\hat{\beta}$
Held-out natural data: $\mathbf{x}\cdot\beta^\star \approx \mathbf{x}\cdot\hat{\beta}$
On accentuation path: $\mathbf{x}\cdot\beta^\star \neq \mathbf{x}\cdot\hat{\beta}$
B
Different optimal penalty
line = DE theory | ○ = MC
noise to signal var. ratio σ²/S=1
Egen / S
pred.
optimal
ridge
Econtrol / S
control
optimal
(0-error)
ridge
Effective penalty κ
C
Prediction invariant
feature transform
Different summary statistics
$F\Sigma F^\top$
$F\Sigma\beta^\star$
E_{gen}
$FF^\top$
$F\beta^\star$
Prediction
(E_{gen})
Control
(N, D)
S, σ², n fixed; σ²/S=.01
Prediction
Control
R²
Feature map F loading on
tail eigenspace (sin²θ)
D
Nonlinear diagnostic of control
Pearson r with control slope in vivo
(site-centered)
log Vφ (neigh.)
log Tφ (trace)
log held out img. MSE
-0.54
-0.61
+0.02
Control slope
(site-centered)
dots =
9 models × 25 sites
log₁₀ Vφ (neigh.)
E
Peer evaluation
N<D
over predicts
N⩾D
under predicts
Control slope
1.35
0.38
slope > 1
in 80/90 in peers
Self-eval
10 cells
Peer review
90 cells